Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years building **high-performance frontend experiences** and **AI-enabled workflows** using React, TypeScript, and Python APIs, focused on speed, reliability, and product impact.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built **real-time, user-facing dashboards** and backend integrations for 500+ users, reducing latency from **1.2s to 350ms** through performant data loading, query optimization, and responsive UI flows.
- Implemented **FastAPI-based AI workflow orchestration** for 5K+ monthly jobs, improving reliability and enabling smoother collaboration between frontend experiences and agent-driven backend processing pipelines.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed **React + Flask** product features handling 10K+ daily requests, creating intuitive interfaces for instructors and integrating LLM-backed grading services into responsive, production-ready user workflows.
- Improved **frontend reliability** and release velocity with CI/CD controls and concurrent processing, cutting turnaround time by **60%** while preserving stable behavior during high-load usage windows.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **Python simulation pipelines** processing 100K+ trajectory logs with reproducible runs, enabling rapid experimentation and data-driven iteration for interactive decision-support and model-validation interfaces.
- Documented debugging outcomes and validation traces, improving cross-team communication speed and reducing friction when translating research outputs into usable product-facing tooling and workflows.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **API and ETL services** with OOP, AWS Glue, and S3, improving response time by **30%** and supporting reliable data flows for feature-rich cross-platform applications.
- Maintained **20+ unit/integration suites** with Jenkins, PyTest, and Mockito across three OS environments, boosting pipeline efficiency by **68%** and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Built **MERN frontend modules** and backend services with REST APIs and WebSockets, supporting eight workflows and 2M+ synchronized records for collaborative, high-availability operations.
- Automated deployment/testing with **Jenkins** and **Docker**, reducing deployment time by **50%** while improving release consistency for rapidly evolving product requirements and stakeholder feedback cycles.

Skills

Languages: **TypeScript**, **JavaScript**, Python, Java, SQL, C++
Frontend: **React.js**, Next.js, responsive UI development, state-driven workflows, performance-focused rendering
Backend & AI: Flask, FastAPI, REST APIs, LLM integration patterns, agentic workflow integration
Cloud & Platform: Kubernetes, Docker, AWS (S3/Glue/RDS), Azure SQL, Git, CI/CD
Engineering Practices: Unit/integration testing, Agile/Scrum, cross-functional collaboration, technical documentation

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built **high-performance frontend interfaces** for 50K+ users and 1M+ transactions with real-time updates, secure integrations, and scalable deployment workflows for latency-sensitive product experiences.

Chat2SQL Code | *Java, TypeScript, SQL, Maven, Docker*
- Developed an **AI-assisted workspace UI** across 10+ databases with schema-aware query generation and structured summaries, enabling faster exploration and human-in-the-loop interaction with LLM-backed services.

HMIS Live | *MERN, ABDM APIs, Twilio*
- Designed **responsive healthcare workflows** supporting 100+ daily appointments, integrating API-backed automation and monitoring checks that improved record retrieval by **2x** and strengthened user-facing reliability.